DIEGO PEDROZA

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science, *Minor in* Intelligent Robotic Systems — *Graduation:* August 2026

Coursework: Data Structures and Algorithms, Computer Logic and Organization, Object-Oriented Programming

Awards: University of Cambridge's AICE Diploma, Florida Academic Scholars Award, UCF's Dean's List S24/F24/S25

SKILLS

Frameworks: TensorFlow, PyTorch, scikit-learn, React, Node.js, FastAPI, Unity, NVIDIA Isaac Sim, ROS/ROS2

Languages: C, C#, C++, Java, Python, Swift, TypeScript, PHP, SQL, Monkey C, Bash

Tools: Linux, REST APIs, web-sockets, Google ADK, Docker, AWS Lightsail, Git, Jupyter Notebook, Jira

WORK EXPERIENCE

UCF's Institute of AI | Undergraduate Research Assistant — **February 2026 – Present**

- Conducting ongoing research in **vision-language-action** robot autonomy, exploring diverse data structures and agentic paradigms to improve task execution through robust language-driven control
- Build **Isaac Sim** extensions for multi-robot environments, exposing **HTTP endpoints** for motion, camera, and IMU sensor data
- Design and test **LLM-based structured-output pipelines** for robot command generation using **Transformers** and **vLLM**
- Execute **GPU-driven** evaluation experiments to establish baselines, analyze experimental results, and inform the quantitative framing of an ongoing robotics paper

Pheratech Systems | Neural Systems Intern — **January 2025 – May 2025**

- Collaborated with a multidisciplinary team to develop robotic vehicles that use decentralized AI and Computer Vision models with real-time data processing and telemetry solutions to enhance autonomous navigation
- Led development of vehicle simulations in **Unity** to validate concepts and create immersive presentations for stakeholders
- Engineered adaptive AI models within an integrated **Unity-Python** framework, applying reinforcement learning techniques and algorithms, such as **PPO**, to optimize performance
- Initiated development of a custom **LLM** aimed at enhancing the interpretation and utilization of complex data

PROJECTS

Polaris – Accessible Indoor Navigation — **January 2026 – Present**

- Advancing Polaris, a deployed iOS indoor navigation platform focused on accessible wayfinding, by prepparing its UCF HEC expansion and supporting future growth across additional buildings
- Integrating **Swift** development, accessibility guidelines, and user experience priorities into routing and design decisions to support inclusive navigation for users with diverse disabilities and mobility needs
- Building a repeatable **Python** and **Flask** CAD/Floor Plan-to-IMDF conversion pipeline to automate floor plan georeferencing, generate **IMDF** mapping assets, and reduce manual mapping and deployment time
- Leading sprint planning and execution as **Project Manager**, applying **Scrum** practices to align team priorities and ensure timely delivery across teams

HotDog - Multi-Agent LLM-Controlled JetBot — **August 2025 – November 2025**

- Collaborated with 4 other engineers to build an autonomous robot on the **NVIDIA Jetson Orin Nano + JetBot platform**, integrating motor control, ultrasonic sensors, and camera
- Established a **multi-agent LLM** command layer (Director / Observer / Pilot) so users could issue vague natural-language goals that were turned into objectives and label sets, enabling generalized behavior without hard-coded routines
- Implemented a **FastAPI + WebSocket** backend to stream **YOLOE** detections and telemetry to a **React/Next.js + Tailwind** frontend for real-time control, monitoring, and goal issuing

Strata: Neuro-Evolution of Augmenting Topologies Implementation — **December 2024 – January 2025**

- Implemented the **NEAT** algorithm from its research paper in **C#** and **Unity**, applying variable topology to simulated agents
- Created "Strata", where agents in diverse environments and identical reward systems evolve different patterns of behavior
- Analyzed the neural networks, learning patterns and emergent behaviors, visualizing data using **Python** and **Matplotlib**

INVOLVEMENT

UCF Theta Tau | Developers Committee Project Manager — **June 2025 – Present**

- Founded the committee to centralize fraternity projects, providing resources and ensuring sustainability
- Directing the development of main projects, such as the chapter website, by defining goals, deadlines, and delegating tasks